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Department of Computer Science and Engineering
Academic Year 2024– 2025 (Odd Semester)

Degree, Semester & Branch	: VII Semester B.E CSE
Course Code & Title	: CCS346 & Exploratory Data Analysis
Name of the Faculty member	: Mrs. S. Vijaya Amala Devi, AP / CSE

Innovative Practice Description

- **Unit / Topic: Unit IV / Bivariate Analysis/** Scatter Plot and Correlations
- **Course Outcome:** CO4
- **Topic Learning Outcome:** 4b
- **Activity Chosen:** Pair Share Repeat
- **Justification:**
  - ❖ To foster deeper understanding of data visualization and correlation analysis through scatterplots, while promoting collaborative learning and critical thinking using the "Pair-Share-Repeat" strategy.
  - ❖ This innovative practice engages students in both independent and team-based analysis, enhancing their communication and analytical skills.
- **Time Allotted for the Activity:** 40 Minutes
- **Procedure for implementation:**
  - ❖ Pair-Share is an effective way of increasing student engagement in the classroom. Asking students to pair-share their answer to a question allows time for students to process their thoughts and discuss it first before sharing it with the class. Students should be encouraged to use complete sentences when responding to a question.
  - ❖ After a pair-share experience, ask students to find a new partner and debrief the wisdom of the old partnership to this new partner.
- **Details of the Implementation:**
  - ❖ Introduce the topic **"Scatterplot and Correlation"** with a thought-provoking question to engage students.
    "What is the relationship between scatterplots and correlation? How can scatterplots help us understand data trends?"
  - ❖ Provide a brief explanation to set the stage:

- Scatterplots are used to visually display the relationship between two variables.
    - Correlation measures the strength and direction of that relationship.

- The class is divided into Pair of students for the first round of discussion. Teams are Anitharani K & Ashwini Bala S, Gokila G & Gomathi Lakshmi M, Haridharshana M U & Kartheeswari M, Mahesh Selvalakshmi M & Jeya Aravinth S, Muthulakshmi M & Praveena A, Selvaruba S & Subhiksha V, Pranav Kumar S & Muthusamy A, Shanmugapriya M & Srivadhani Karthiga A, Subana Devi G & Swathi K, Swetha Sri S & Thangaselvi S, Thilothamai K & Sundara Mahalingam P and Vignesh Prabhu M and Vishnu A.
- Students discuss the question with their assigned partners.
- Instructions for students:
    - Student A explains their understanding of **scatterplots** for 2 minutes.
    - Student B shares their understanding of **correlation** for 2 minutes.
- Encourage students to:
    - Use examples to connect scatterplots and correlation (e.g., "positive/negative trends").
    - Speak in complete sentences when explaining.
- Time Limit is 5 minutes for the first round.
    - *Anitharani K:* "Scatterplots show how two variables relate. If the points trend upward, we see a positive relationship."
    - *Ashwini Bala S:* "Correlation measures this relationship. A strong positive correlation means the variables increase together."
- Instruct students to find new partners for the next round.

- New pairs can be formed.
- Students summarize the key insights or "wisdom" they learned from their first partner to their new partner.
- Instructions for students:
    - Share one example or point their previous partner discussed.
    - Add their own reflections or understanding of the topic.
    - Time Limit **is** 5 minutes for the second round.
- Invite volunteers to share insights from their pair discussions with the class.
- Example prompts:
    - "What is something interesting or new your partner shared?"
    - "How did discussing scatterplots and correlation with two partners improve your understanding?"
    - "I learned that scatterplots help identify trends visually, while correlation gives us a number to explain that trend."
    - "My partner gave an example of height and weight having a positive correlation, which I hadn't thought of before."

CO – PO / PSO mapping:

CO	PO1	PO2	PO3	PO9	PO10	PO12	PSO3
CO4	3	3	3	3	3	3	3

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO9	PO10	PO12
	3	**3**	**3**	**2**	**2**	**3**
Justification for correlation	Students engage in comprehensive data analysis, using scatterplots and correlation to apply their engineering knowledge effectively.	Students develop skills to identify, formulate, and analyze complex problems	Students contribute to the design of data-driven solutions, demonstrating how to approach and solve engineering problems creatively and collaboratively	Students enhance their ability to work both independently and in teams.	Written communication/ Asking questions are improved	Student will implement the questioning skillin future in written format.

- **Images / Screenshot of the practice:**

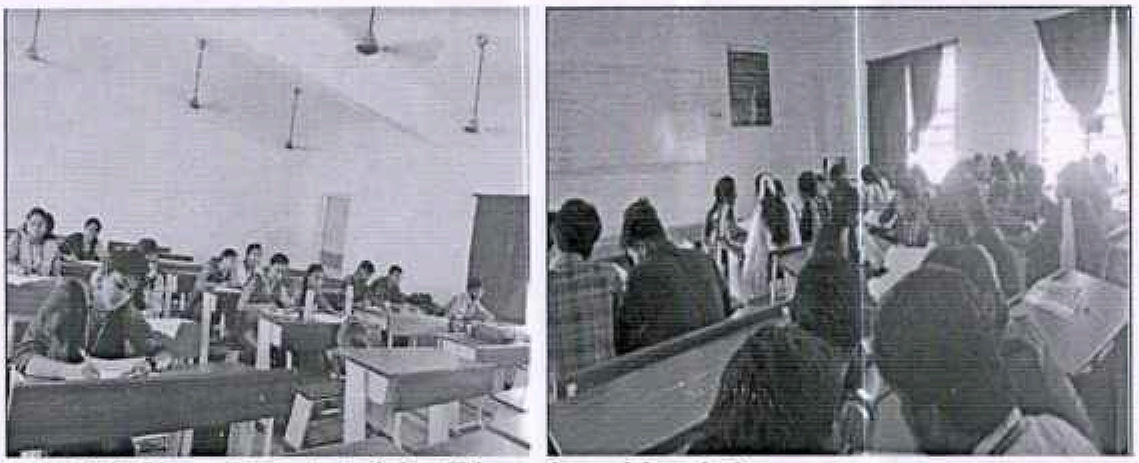

Figure 1: Pair Share Repeat Activity (Discussing with pairs)

Reflective Critique:

- ***Discussing with students:***
  - Prompts students to analyze and evaluate the information presented, fostering deeper intellectual engagement.

- ***Benefit of the practice:***
  - Encourages immediate application of concepts, promoting better retention.
  - Fosters teamwork and idea exchange, exposing students to diverse perspectives.
  - Develops questioning, critique, and communication skills essential for professional growth.

- Students learn to articulate thoughts clearly using complete sentences during pair-share activities.
- Enhances both verbal and written communication as they express and debrief ideas.
- New partnerships allow them to rephrase and refine thoughts.

❖ ***Challenges faced in implementation:***

- Some students hesitated to actively contribute.
- A few students lacked familiarity with the dataset, limiting their initial involvement.
- Pair-share activities may take up significant class time, potentially reducing time for other learning tasks.
- Lack of proper monitoring can lead to inaccuracies.
- Pairing students effectively can be challenging, especially if there are personality clashes or communication gaps.

References:

- **https://www.aisnsw.edu.au/teachers-and-staff/teaching-and-learning/literacy-and-numeracy/foundations-of-effective-instruction/engagement-strategies/pair-share**
- **https://my.chartered.college/research-hub/think-pair-share-a-three-step-collaborative-learning-strategy-that-helps-pupils-consider-questions-in-more-depth/**

19/12/24
Signature of Faculty Member

19/12/2024
HOD